

ARGHYADIP PAKHIRA

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Arghyadevs

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Overview

B.Tech CSE student specializing in full-stack development and AI/ML, with hands-on experience building production-grade applications using RAG pipelines, vector search, and cloud infrastructure.

Education

Year	Degree / Exam	Institute	Score
2023 – Present	B.Tech in Computer Science & Engineering	Techno Main Salt Lake	7.4 / 10
2022	WBCHSE (Class XII)	Narajole Mahendra Academy	96.8%
2020	WBBSE (Class X)	Narajole Mahendra Academy	90.6%

Coursework / Skills

Core Subjects: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

Languages: C++, Python, JavaScript, Swift, SQL

Technologies: React.js, Next.js, Node.js, FastAPI, Flask, MongoDB, ChromaDB, Cloudinary

Tools: Git, Docker, Linux, AWS (EC2, S3, CloudFront), Vite

ML / AI: TensorFlow, PyTorch, OpenCV, YOLO, FAISS, Scikit-learn, LangChain, Transformers, Edge-TTS

Projects

NeuralFlix – Semantic Movie Recommendation Engine2025

- Indexed **1,048 titles** (623 movies + 425 TV series) using **768-dimensional transformer embeddings** (all-mpnet-base-v2) stored in FAISS, achieving **sub-50ms** nearest-neighbor search latency at scale.
- Replaced keyword-based search with semantic vector retrieval, surfacing contextually relevant results for abstract or descriptive queries that traditional search engines fail to handle.
- Built a responsive **Next.js 14** frontend with real-time query performance, displaying cosine similarity-based match scores to surface ranked results with transparent relevance indicators.
- Implemented genre-aware metadata filtering layered on top of vector search, enabling hybrid queries that combine semantic intent with structured constraints (e.g., genre, release year).

creAItr – Unified AI Creative Workspace2025

- Architected a unified AI creative suite integrating **6 specialized engines** (RAG-powered Chat, Cinematic Editor, Photo Editor, Neural Canvas, Writing Studio, Asset Vault) supporting **5+ rich media formats**.
- Engineered a local **RAG pipeline** using **ChromaDB** and Flask, achieving **sub-500ms** retrieval latency across PDFs and live RSS feeds while ensuring **100% data privacy** with zero external API calls.
- Built an immersive **React 19 / Three.js** frontend on **AWS CloudFront**, maintaining **60 FPS** 3D rendering via LOD geometry; integrated **Llama-3.1-70B** and **Edge-TTS** for a **500+ concurrent user** capable backend.
- Designed a modular plugin architecture allowing each creative engine to operate independently or in tandem, enabling seamless cross-module asset sharing (e.g., generated images passed directly into the Cinematic Editor).

AlgoGuard AI – Algorithmic Bias Detection Platform2025

- Engineered an end-to-end ML pipeline to detect and mitigate algorithmic bias in loan approval models, reducing **Statistical Parity Difference (SPD)** from **0.38 to 0.05** – an **85% fairness improvement**.
- Benchmarked **3 bias mitigation algorithms** (Feature Removal, Reweighting, SMOTE Re-sampling) across demographic sub-groups while maintaining **92% predictive accuracy** post-mitigation.
- Developed a high-performance **C++ graph analytics engine** to compute structural similarity networks, visually exposing demographic discrimination patterns in model decision boundaries.

Achievements & Certifications

- LeetCode Top 4.2% globally** – solved **1200+** problems spanning dynamic programming, graphs, and system design.
- 4-Star HackerRank** in Problem Solving – consistently ranked in competitive programming contests.
- National Finalist, Smart India Hackathon (SIH)** – selected among top teams across India out of thousands of submissions.
- HackerRank Certified** – Problem Solving Intermediate (Credential ID: C62D2373E9F0).
- GFG Certified** – iOS Development for Beginners: Master Swift & App Building (May 2025).
- Open Source Contributor** – GirlScript Summer of Code (GSSoC) Extended 2024; contributed to multiple repositories in frontend and ML tooling.

Positions of Responsibility

- Campus Lead & Campus Mantri, GeeksforGeeks** – organized weekly coding sessions, curated structured DSA learning paths, mentored peers on placement preparation, and managed community engagement for the college chapter.
- Core Member & Sponsorship Head, Geekonix Tech Club** – led sponsor outreach and event logistics for college-level technical and cultural events; conducted hands-on workshops in competitive programming, DSA, and full-stack development, mentoring junior students from foundations to deployment.